

Education

University of Maryland, College Park

PhD., Computer Science

COLLEGE PARK, MD, USA

Aug 2018 – Oct 2023

Advisor: Prof. David Jacobs

Thesis: Self-Supervised Learning for Large Scale Datasets.

MS, Computer Science

Aug 2018 – Aug 2020

Birla Institute of Technology, Mesra

Bachelor of Engineering (BE), Computer Science

RANCHI, JHARKHAND, INDIA

Jun 2012 – Jun 2016

Publications

Published

- [1]Shlok Mishra*, Songwei Ge*, Chun-Liang Li, Simon Kornblith, David Jacobs.
"Hyperbolic Contrastive Learning for Visual Representations beyond Objects." *CVPR 2023*
- [2]Songwei Ge, Shlok Mishra, Chun-Liang Li, Haohan Wang, David Jacobs.
"Robust Contrastive Learning Using Negative Samples with Diminished Semantics." *NeurIPS 34*
- [3]Shlok Mishra, Anshul Shah, Ankan Bansal, Abhyuday Jagannath, Abhishek Sharma, David Jacobs, Dilip Krishnan.
"Object-Cropping for Self-Supervised Learning." *TMLR 2022; CoLLAs 2023*
- [4]Tianhong Li, Huiwen Chang, Shlok Mishra, Han Zhang, Dina Katabi, Dilip Krishnan.
"MAGE: MASKed Generative Encoder to Unify Representation Learning and Image Synthesis." *CVPR 2023*
- [5]Anshul Shah, Aniket Roy*, Ketul Shah*, Shlok Mishra, David Jacobs, Anoop Cherian, Rama Chellappa.
"HaLP: Hallucinating Latent Positives for Skeleton-based Self-Supervised Learning of Actions." *CVPR 2023*
- [6]Shlok Mishra, Kuntal Sengupta, Max Horowitz-Gelb, Vincent Chu, Sofien Bouaziz, David Jacobs.
"Improved Presentation Attack Detection Using Image Decomposition." *IJCB 2022 (Oral)*
- [7]Shlok Mishra, Anshul Shah, Ankan Bansal, Jonghyun Choi, Abhinav Shrivastava, Abhishek Sharma, David Jacobs.
"Learning Visual Representations for Transfer Learning by Suppressing Texture." *BMVC 2022*
- [8]Anshul Shah, Shlok Mishra, Ankan Bansal, Jun-Cheng Chen, Rama Chellappa, Abhinav Shrivastava.
"Pose and Joint-Aware Action Recognition." *WACV 2022*
- [9]Janit Anjaria, Hong Wei, Shlok Mishra, Hanan Samet.
"TrajDistLearn: Learning To Compute Distance Between Trajectories." *ACM SIGSPATIAL*
- [10]Shlok Kumar Mishra, Pranav Kumar, Sujana Kumar Saha.
"A Support Vector Machine Based System for Technical Question Classification." *In MIKE 2015*
- [11]Aditya Sarkar, Yi Li, Jiacheng Cheng, Shlok Kumar Mishra, Nuno Vasconcelos.
"Leveraging Data to Say No: Memory Augmented Plug-and-Play Selective Prediction." *ICLR 2026*
- [12]Xuanming Cui, Jianpeng Cheng, Hong-you Chen, Satya Narayan Shukla, Abhijeet Awasthi, Xichen Pan, Chaitanya Ahuja, Shlok Kumar Mishra, Yonghuan Yang, Jun Xiao, Qi Guo, Ser-Nam Lim, Aashu Singh, Xiangjun Fan.
"Think then embed: Generative context improves multimodal embedding" *ICLR 2026*
- [13]Meta AI.
"The Llama 4 Herd: The Beginning of a New Era of Natively Multimodal AI Innovation." *Technical Report, 2025*
- [14]Xichen Pan, Aashu Singh, Satya Narayan Shukla, Xiangjun Fan, Shlok Kumar Mishra, Saining Xie.
"RepFusion: Leveraging Multimodal Priors for Denoising in Representation Space." *NeurIPS 2026*
- [15]Chao Li, Tianhong Li, Sai Vidhyaranya Nuthalapati, Hong-You Chen, Satya Narayan Shukla, Jianpeng Cheng, Yonghuan Yang, Jun Xiao, Xiangjun Fan, Aashu Singh, Dina Katabi, Shlok Kumar Mishra.
"Unifying Contrastive and Generative Objectives for Visual Understanding and Text-to-Image Generation." *NeurIPS 2026*
- [16]Aditya Sarkar, Yi Li, Zihao Wang, Jiacheng Cheng, Sai Vidhyaranya Nuthalapati, Aashu Singh, Shlok Kumar Mishra, David Jacobs, Nuno Vasconcelos.
"An Attribute-Based Measure of Video Complexity." *CVPR 2026 Workshop*
- [17]Rajeev Bhatt Ambati, Tianyi Niu, Aashu Singh, Shlok Mishra, Snigdha Chaturvedi, Shashank Srivastava.
"Socratic Students: Training Language Models to Ask Better Questions for Reasoning." *Findings of ACL*

Submissions

- [18]Guangyu Sun, Shlok Kumar Mishra, Wentao Bao, Robert Zhenheng Yang, Xiao Wang, Xiyuan Wang, Yujunrong Ma, Chen Yuan, Max Xiangjun Fan, Jun Xiao, Jianpeng Cheng.
"FLAT: Resampling Image and Text into 1D Flexible-Length Aligned Transmodal Tokens for Retrieval and Generation."
- [19]Xuanming Cui, Shlok Kumar Mishra, Wentao Bao, Aashu Singh, Zihao Wang, Xiangjun Fan, Jun Xiao, Ser-Nam Lim, Jianpeng Cheng.
"MoEMB: Scaling Universal Multimodal Embeddings with Efficient Mixture-of-Experts Models."
- [20]Jianpeng Cheng, Xian Wu, Jiangfan Zhang, Wentao Bao, Chaitanya Ahuja, Shlok Kumar Mishra, Xuanming Cui, Hanchao Yu, Yang Gao, Fan Xia, Haixing Dai, Frankie Yuan, Zihao Wang, Xiaobing Chen, Qi Guo, Shaodan Zhai, Aashu Singh, Xiangjun Fan, Jun Xiao.
"TTE-Flash: Accelerating Reasoning-based Multimodal Representations via Think-Then-Embed Tokens."
- [21]Xichen Pan, Satya Narayan Shukla, Aashu Singh, Zhuokai Zhao, Shlok Kumar Mishra, Jialiang Wang, Zhiyang Xu, Jiuhai Chen, Kunpeng Li, Felix Juefei-Xu, Ji Hou, Saining Xie.
"Transfer between Modalities with MetaQueries."
- [22]Xichen Pan, Satya Narayan Shukla, Aashu Singh, Zhuokai Zhao, Shlok Kumar Mishra, Jialiang Wang, Zhiyang Xu, Jiuhai Chen, Kunpeng Li, Felix Juefei-Xu, Ji Hou, Saining Xie.
"Render unto Diffusion! Keep LLMs Frozen for Unified Multimodal Modeling."

[23]Yanlai Yang, Zhuokai Zhao, Satya Narayan Shukla, Aashu Singh, Shlok Kumar Mishra, Lizhu Zhang, Mengye Ren.
"StreamMem: Query-Agnostic KV Cache Memory for Streaming Video Understanding."

[24]Xuanming Cui, Hong-you Chen, Hao Yu, Hao Yuan, Zihao Wang, Shlok Kumar Mishra, Hanchao Yu, Yonghuan Yang, Jun Xiao, Ser-Nam Lim, Jianpeng Cheng, Qi Guo, Xiangjun Fan.

"Reason to Contrast: A Cascaded Multimodal Retrieval Framework."

Arxiv

[25]Shlok Mishra, Tsung-Yu Lin, Linda Wang, Hongli Xu, Yimin Liu, Michael Hsu, Chaitanya Ahuja, Hao Yuan, Jianpeng Cheng, Hong-You Chen, Haoyuan Xu, Chao Li, Abhijeet Awasthi, Jihye Moon, Don Husa, Michael Ge, Sumedha Singla, Arkabandhu Chowdhury, Phong Dinh, Satya Narayan Shukla, Yonghuan Yang, David Jacobs, Qi Guo, Jun Xiao, Xiangjun Fan, Aashu Singh.

"Xray-Visual Models: Scaling Vision models on Industry Scale Data." *Demo at Meta booth (with DINO and MetaCLIP)*

[26]Shlok Mishra*, Joshua David Robinson*, Huiwen Chang, David Jacobs, Aaron Maschinot, Dilip Krishnan.

"CAN: A Simple Contrastive Masked Autoencoder Framework For Scaling To Uncurated Data."

[27]Shlok Kumar Mishra, Pranav Goel, Abhishek Sharma, Abhyuday Jagannath, David Jacobs, Hal Daumé III.

"Question Generation for Longer Sequences." <https://arxiv.org/pdf/2004.05109.pdf>

Professional Experience

Meta AI – Super Intelligence Team

MENLO PARK, CA

Senior AI Research Scientist

Jun '25 – Present

- Tech lead for a team of five researchers developing large-scale vision encoders (CLIP, SimCLR, DINO, MAE) for LLaMA and also serving 200+ internal clients.
- Leading the team for post-training of MMLLM with image/video generation.
- Building efficient, large-scale MMLLMs for Meta-scale traffic; training vision encoders and multimodal models leveraging LLaMA.
- Host and mentor 5 research interns, collaborating on academic papers in the areas of LLMs and large-scale computer vision.

AI Research Scientist

Mar '24 – May '25

- Worked on large-scale pre-training of LLM and vision encoder.

Google DeepMind

CAMBRIDGE, MA

Research Intern

Aug '21 – May '23

- Mentor: Dr. Dilip Krishnan.
- Self-supervised learning for uncurated data; combined generative and discriminative features to improve recognition and generation.
- **Product Impact:** CAN embeddings explored for segmentation, detection, and adversarial perturbations. Two Google Lens groups observed a 13% error-rate reduction.

Google Research

SEATTLE, WA

Research Intern

May '23 – Sep '23

- Mentor: Prof. Ira Kemelmacher-Shlizerman.
- Worked on TryOn Diffusion; added controllability and mitigated overfitting in try-on models.

Google Augmented Reality

MOUNTAIN VIEW, CA

Research Intern

May '21 – Aug '21

- Mentors: Dr. Christian Haene, Dr. Hossam Isack.
- Domain adaptation with GANs using inference-via-optimization for controllable outputs.
- Proposed a StyleGAN control architecture using additional keypoint information.

Google DeepMind

MOUNTAIN VIEW, CA

Research Intern

May '20 – Jan '21

- Mentors: Dr. Kuntal Sengupta, Dr. Vincent Chu, Dr. Sofien Bouaziz.
- Face anti-spoofing via intrinsic image properties; showed albedo outperforms depth for spoof detection.
- Internship work accepted as *Oral* at a top-tier vision/biometrics venue.
- **Product Impact:** Ideas contributed to Pixel 7 Face Unlock deployment.

Flipkart Internet Pvt. Ltd.

BANGALORE, INDIA

Software Development Engineer

Dec '16 – Aug '18

- Address-to-zipcode prediction using LSTM/BiLSTM, SVM, multinomial logistic regression, ridge regression, LDA, and LSI.
- Vehicle Routing: Java Spring Boot + OptaPlanner end-to-end route planner; added slotted delivery and load balancing with Google OR-Tools.
- Dynamic geo-clustering using K-means and DBSCAN to boost last-mile productivity.

Selected Achievements

- Doctoral Consortium, ICCV 2023 (mentors: Amir R. Zamir & Jakob Verbeek).
- Doctoral Consortium, BMVC 2023.
- MAGE featured by MIT News.
- Goldhaber Travel Grant and CS Dept Travel Grant (UMD) for ICCV.
- Invited talk at Google PIRC (PhD Internal Research Conference).
- Outstanding reviewer in ECCV 2026.

Service

Serving as reviewer for major computer vision and ML venues:

CVPR [20–26], NeurIPS [20–26], ICLR [21–26], AAAI [21–26], ECCV [20, 22, 24, 26], ICCV [21, 23, 25], ICML [21–26], WACV [21–26], BMVC [22–23];
NLP venues: ACL [2020–2022], EMNLP [2020–2022];

Journals: TPAMI, CVIU, TMLR, IJCV, TIP, IEEE-TCSVT, IEEE-IFS, etc.

Interns and Academic Collaborators

- **Charlie Lie** (MIT) – Advisor: Prof. Dina Katabi
 - **Xichen Pan** (NYU) – Advisor: Prof. Saining Xie
 - **Yanlai Yang** (NYU) – Advisor: Prof. Mengye Ren
 - **Xuanming Cui** (University of Central Florida) – Advisor: Prof. Ser-Nam Lim
 - **Aditya Sarkar** (University of Maryland)
 - **Fancong Yin** (NYU) – Advisor: Prof. Marten van Schijndel
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Teaching Experience

- Teaching Assistant, CMSC828L Deep Learning (Spring 2021, David Jacobs).
 - Teaching Assistant, CMSC 420 Data Structures (Fall 2018, Hanan Samet).
 - Teaching Assistant, Computer Vision (Spring 2019, Cornelia Fermüller).
 - Mentored high school students (Summer 2019) on LeafSnap project implementation.
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Skills

Computer Vision, Machine Learning, Natural Language Processing, Deep Learning, Matlab, Java, PyTorch, TensorFlow, JAX, Python.
